Appendix

A. Oscillator Strengths, Radiative Rates, and Excitation Energies for Hydrogen and Helium

A.1 Oscillator Strengths and Radiative Rates for Hydrogen

Table A.1 Oscillator strengths (f_{nm}) and radiative transition probabilities (A_{mn}) for hydrogen.

Transition $n\ell \to m\ell'$	f_{nm}	$A_{mn}[s^{-1}]$	Transition $n\ell \to m\ell'$	f_{nm}	$A_{mn}[s^{-1}]$
1s-2p	0.4162	6.265×10^{8}	3s-4p	0.4847	3.065×10^{6}
1s-3p	7.910×10^{-2}	1.672×10^{8}	3s-5p	0.1210	1.638×10^{6}
1s-4p	2.899×10^{-2}	6.818×10^{7}	3s-6p	5.139×10^{-2}	9.551×10^{5}
1s-5p	1.394×10^{-2}	3.437×10^{7}			
1s-6p	7.800×10^{-3}	1.973×10^{7}	3p-4s	3.225×10^{-2}	1.835×10^{6}
			3p-5s	7.428×10^{-3}	9.046×10^{5}
2p-3s	1.359×10^{-2}	6.313×10^{6}	3p-6s	3.032×10^{-3}	5.071×10^{5}
2p-4s	3.045×10^{-3}	2.578×10^{6}			
2p-5s	1.213×10^{-3}	1.289×10^{6}	3p-4d	0.6183	7.037×10^{6}
2p-6s	6.180×10^{-4}	7.350×10^{5}	3p-5d	0.1392	3.391×10^{6}
			3p-6d	5.614×10^{-2}	1.878×10^{6}
2s-3p	0.4349	2.245×10^{7}			
2s-4p	0.1028	9.668×10^{6}	3d-4p	1.099×10^{-2}	3.475×10^{5}
2s-5p	4.193×10^{-2}	4.948×10^{6}	3d-5p	2.210×10^{-3}	1.495×10^{5}
2s-6p	2.163×10^{-2}	2.858×10^{6}	3d-6p	8.420×10^{-4}	7.824×10^{4}
2p-3d	0.6958	6.465×10^{7}	3d-4f	1.018	1.379×10^{7}
2p-4d	0.1218	2.062×10^{7}	3d-5f	0.1566	4.542×10^{6}
2p-5d	4.437×10^{-2}	9.425×10^{6}	3d-6f	5.389×10^{-2}	2.146×10^{6}
2p-6d	2.163×10^{-2}	5.145×10^{6}			

Reference: Wiese et al. (1966)

A.2 Oscillator Strengths Averaged over Angular Momentum Quantum Number for Hydrogen

Table A.2. Oscillator strengths (f_{nm}) and radiative transition probabilities (A_{mn}) averaged over angular momentum quantum numbers, for hydrogen.

Transition n→m	f_{nm}	$A_{mn}[s^{-1}]$	Transition n→m	f_{nm}	$A_{mn}[s^{-1}]$
1-2(L_α)	0.4162	4.699×10^{8}	3-4(P_α)	0.8421	8.986×10^{6}
1-3(L_β)	7.910×10^{-2}	5.575×10^{7}	3-5(P_β)	0.1506	2.201×10^{6}
1-4(L_γ)	2.899×10^{-2}	1.278×10^{7}	3-6(P_γ)	5.584×10^{-2}	7.783×10^{5}
1-5(L_δ)	1.394×10^{-2}	4.125×10^{6}	3-7(P_δ)	2.768×10^{-2}	3.358×10^{5}
1-6(L_ε)	7.799×10^{-3}	1.644×10^{6}	3-8(P_ε)	1.604×10^{-2}	1.651×10^{5}
2-3(H_α)	0.6407	4.410×10^{7}	4-5	1.038	2.699×10^{6}
2-4(H_β)	0.1193	8.419×10^{6}	4-6	0.1793	7.711×10^{5}
2-5(H_γ)	4.467×10^{-2}	2.530×10^{6}	4-7	6.549×10^{-2}	3.041×10^{5}
2-6(H_δ)	2.209×10^{-2}	9.732×10^{5}	4-8	3.230×10^{-2}	1.424×10^{5}
2-7(H_ε)	1.270×10^{-2}	4.389×10^{5}	4-9	1.870×10^{-2}	7.459×10^{4}

Reference: Wiese et al. (1966)

Johnson formula for f_{nm}:

$$f_{nm}(m>n) = \frac{32}{3(\sqrt{3})\pi}\,\frac{n}{m^3}\,\frac{1}{x^3}\,g(n,x),$$

$$x = 1 - \left(\frac{n}{m}\right)^2,$$

$$g(n,x) = g_0(n) + g_1(n)\,\frac{1}{x} + g_2(n)\,\frac{1}{x^2}\,.$$

Table A.3. Coefficients in the Johnson formula.

n =	1	3	≥ 3
g_0	1.1330	1.0785	$0.9935 + 0.2328\ n^{-1} - 0.1296\ n^{-2}$
g_1	-0.4059	-0.2319	$-n^{-1}(0.6282 - 0.5598\ n^{-1} + 0.5299\ n^{-2})$
g_2	+0.07014	0.02947	$n^{-2}(0.3887 - 1.181\ n^{-1} + 1.470\ n^{-2})$

Reference: Johnson (1972)

A.3 Ionization and Excitation Energies of Low-Lying States of Helium

Table A.4. Ionization and excitation energies of low-lying states lsnℓ of helium, with n=1, 2, 3, 4.

State	Singlets, ^1L		Triplets, ^3L	
lsnℓ	$E^{exc}_{n\ell}$[eV]	$E^{ion}_{n\ell}$[eV]	$E^{exc}_{n\ell}$[eV]	$E^{ion}_{n\ell}$[eV]
1s2	24.588			
1s2s	20.614	3.973	19.818	4.769
1s2p	21.217	3.371	20.963	3.625
1s3s	22.919	1.669	22.717	1.871
1s3p	23.086	1.502	23.006	1.582
1s3d	23.073	1.515	23.072	1.515
1s4s	23.672	0.9155	23.529	0.9951
1s4p	23.741	0.8470	23.706	0.8812
1s4d	23.735	0.8529	23.735	0.8530
1s4f	23.736	0.8520	23.736	0.8520

For n = 5, 6, and 7 we use an average over angular momentum and total spin.

Table A.5. Ionization and excitation energies of states of helium with n ≥ 5.

State n	$\langle E^{exc}_{n\ell}\rangle_{\ell,spin}$	$\langle E^{ion}_{n\ell}\rangle_{\ell,spin}$
5	24.07	0.52
6	24.71	0.37
7	24.30	0.29
≥8	$E^{ion}_{1s^2} - \frac{Ry}{n^2}$	Ry/n^2

Based on Wiese et al. (1966).

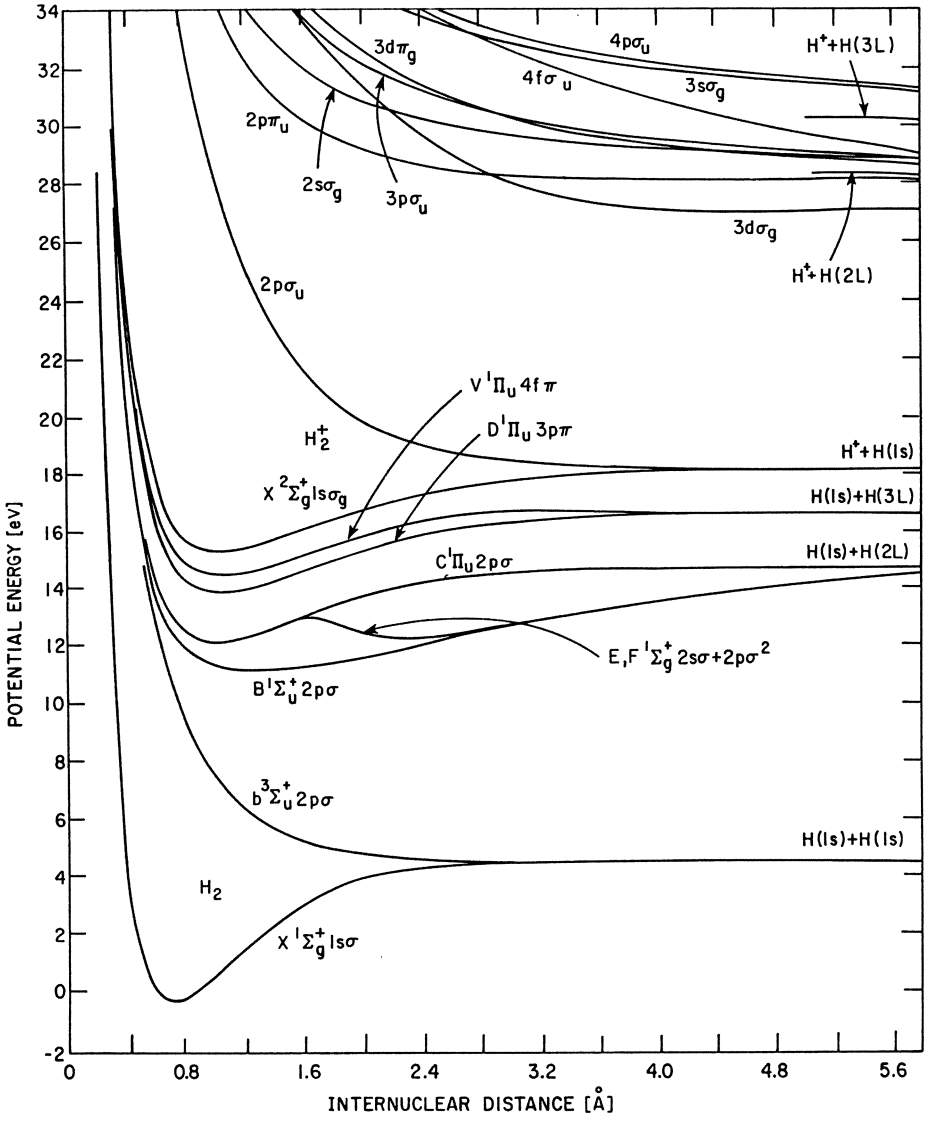
A.4 Oscillator Strengths and Radiative Rates for Helium

Table A.6. Oscillator strengths (f_{nm}) and radiative transition probabilities (A_{mn}) for allowed transitions in helium

Transition		f_{nm}	$A_{mn}[s^{-1}]$
$1s^2$-1s2p	^1S-$^1P^o$	0.276	1.80×10^9
-1s3p	^1S-$^1P^o$	0.0734	5.66×10^8
-1s4p	^1S-$^1P^o$	0.0302	2.46×10^8
-1s5p	^1S-$^1P^o$	0.0153	1.28×10^8
1s2s-1s2p	^1S-$^1P^o$	0.376	1.98×10^6
-1s3p	^1S-$^1P^o$	0.151	1.34×10^7
-1s4p	^1S-$^1P^o$	0.0507	0.72×10^7
-1s5p	^1S-$^1P^o$	0.0221	3.76×10^6
1s2s-1s2p	^3S-$^3P^o$	0.300	1.02×10^7
^3S-$^3P^o$	0.180	1.02×10^7	
^3S-$^3P^o$	0.060	1.02×10^7	
1s2s-1s3p	^3S-$^3P^o$	0.0645	0.95×10^7
-1s4p	^3S-$^3P^o$	0.0231	5.05×10^6
-1s5p	^3S-$^3P^o$	0.0114	2.93×10^6
1s2p-1s3s	$^1P^o$-^1S	0.0480	1.81×10^7
-1s4s	$^1P^o$-^1S	0.834×10^{-2}	0.65×10^7
-1s5s	$^1P^o$-^1S	0.308×10^{-2}	3.13×10^6
1s2p-2s3d	$^1P^o$-^1D	0.711	0.64×10^8
-1s4d	$^1P^o$-^1D	0.122	2.02×10^7
-1s5d	$^1P^o$-^1D	0.0436	0.91×10^7
1s2p-1s3s	$^3P^o$-^3D	0.9693	1.54×10^7
-1s3s	$^3P^o$-^3S	0.0692	0.93×10^7
-1s4s	$^3P^o$-^3S	0.0118	1.06×10^7
-1s5s	$^3P^o$-^3S	0.365×10^{-2}	4.30×10^6
1s2p-1s3d	$^3P^o$-^3D	0.609	0.70×10^8
-1s4d	$^3P^o$-^3D	0.125	2.51×10^7
-1s5d	$^3P^o$-^3D	0.0474	1.17×10^7

Reference: Wiese et al. (1966)

B. Potential Energy Diagram for H_2 and H_2^+



The potential energy curves for hydrogen; see Sharp (1972).

C. Values of the Function $D(\beta)$

$\beta < 10^{-3}$: $D(\beta) = 4\beta \ln(1.4/\beta)$,

$\beta > 10$: $D(\beta) = \frac{\beta}{2} \exp(-\sqrt{2\beta})$,

$10^{-3} \leq \beta \leq 10$: $D(\beta)$ is given in Table C.1.

Table C.1. Values of $D(\beta)$

$1/\beta$	0.1	0.2	0.3	0.4	0.5	0.6	0.8	1.0	1.25	1.5	2.0	2.5
$D(\beta)$	0.057	0.104	0.135	0.157	0.175	0.194	0.230	0.264	0.296	0.328	0.367	0.388
$1/\beta$	3.0	3.5	4.0	4.5	5	6	8	10	12.5	15	20	25
$D(\beta)$	0.399	0.405	0.410	0.405	0.399	0.380	0.345	0.318	0.285	0.263	0.227	0.205
$1/\beta$	30	40	60	80	100	150	200	300	400	600	800	1000
$D(\beta)$	0.190	0.168	0.141	0.124	0.110	0.092	0.080	0.064	0.054	0.042	0.035	0.028

Reference: Janev and Presnyakov (1980)

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